SimpleTracking Instructions

SimpleTracking Version 1.0

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Step 0: Adjusting Videos to work in SimpleTracking

See Kailie's tutorial here to learn how to convert videos to image files and parameter settings.

Step 1: SimpleTracking Parameters

1. Open MATLAB and open `SimpleTracking.m`
2. Adjust Main parameters. <u>The most important parameters</u> are `numFrames` and `secondVideo`. Make sure those two parameters are correct.

```
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%% MAIN PARAMETERS

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
bpfiltsize=         20;                         % bpfiltsize        [A]
hthreshopt=         '2. Std. Thresh.';          % Threshold Option  [B]
defaultthresh2=     4.0;                        % thr (>1)          [C]

stdevMinThreshold=  2;                          % Cull: Std. Min. Threshold
numFrames=          200;                        % Cull: # of Video Image Frames
secondVideo=        10;                         % Cull: Length of Video (sec)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```
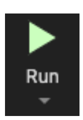

3. Adjust other parameters if necessary. All the parameters are pre-set to the setting that is frequently used in the lab. See Kailie's tutorial or TrackingGUI manual if you are unsure which parameters you should use.

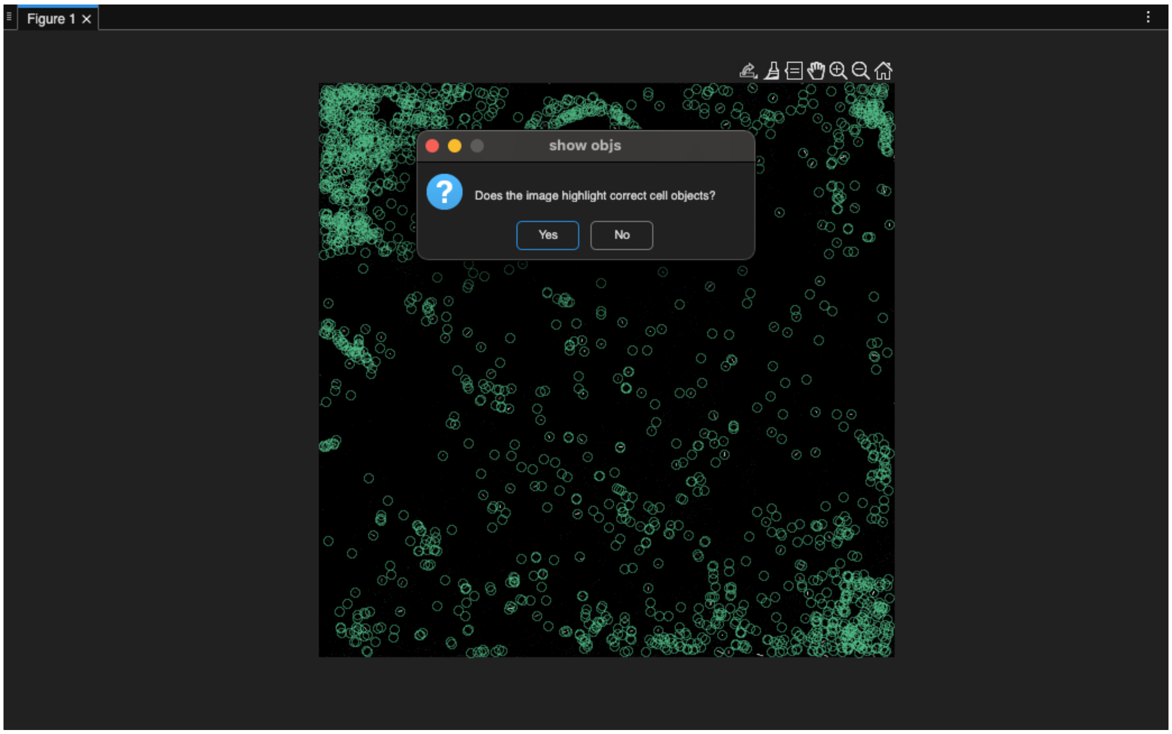
Step 2: Select Images

1. Click Run 

2. The script will prompt you to select the image files. Browse to the folder with the images
3. Select the first image (format should be `.tiff`)
4. Select the last image

Step 3: Confirm Objects

1. The first image of the video will pop up, with green circles on each bacterium object to show that the software will track them.
2. Click "Yes" if the objects look correct. Otherwise, click "No" and the software will stop. Adjust parameters and re-run if incorrect.



Step 4: Track, Link and Cull Objects

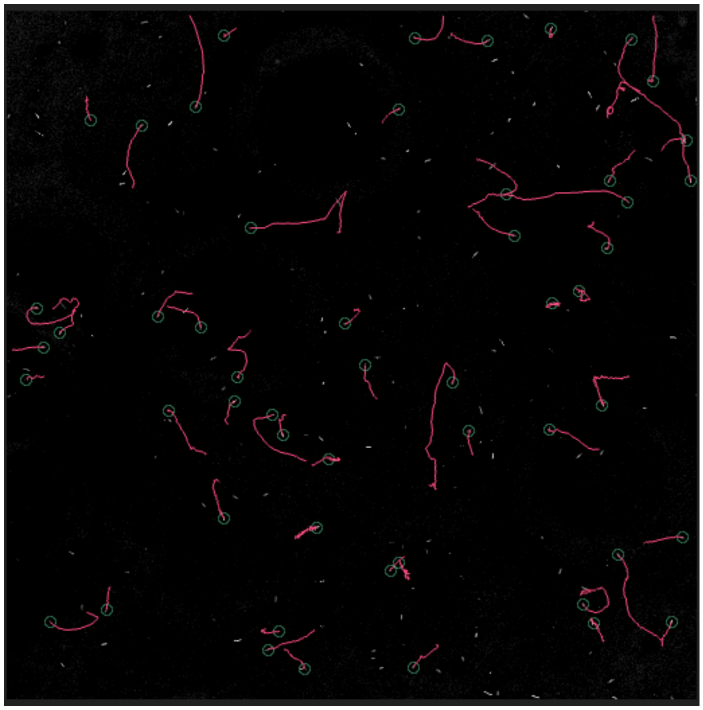
1. After confirming, the software will proceed with tracking, Linking and Culling all objects. You can see the progress with the progress bar.
2. If parameters aren't changed, the software will cull twice (`Stdev`, `TrackLength`)
3. You'll be able to see the number of tracks after culling on the Command Window. Copy the number if needed. In the screenshot, 163996 is # tracks before the cull and 340 is after.

```
>> SimpleTracking
Found 163966 unique tracks.
More than 10 tracks found; not displaying details.
Number of Tracks After Cull:
   340
```

4. I've noticed that the number of tracks after cull is larger in SimpleTracking than TrackingGUI. I believe that it is because TrackingGUI tracks a subset of images (what it shows on the GUI) while simpleTracking uses the entire image without cropping.

Step 5: Save Objects

1. Next, another popup will show with the object tracks after the cull with red lines. Confirm that the tracks look correct.



2. SimpleTracking saves the output and automatically runs Kailie's additional scripts and `migration.m` afterwards.
3. There will be two outputs:
   a. `SimpleTrackingoutput.mat`
   b. `Tracked_positions_firstSecond.xlsx`

Context and References

SimpleTracking is a MATLAB script written for Baylink Lab (www.baylink-lab.com) to simplify bacteria tracking from videos and provide bacteria movement metrics, utilizing TrackingGUI scripts consolidated with other scripts written by Baylink lab members. Please refer to the links below for further information:

- TrackingGUI GitHub: https://github.com/rplab/TrackingGUI_and_Localization_Public
- ImageJ and TrackingGUI instructions by Kailie Franco: https://github.com/clara-h-shin/SimpleTracking/blob/main/swim_tracking_tutorial_by_kailie.docx